RAINFALL ANALYSIS

Analysis and Visualization of Historical Rainfall Patterns in India Using Python

Project Report

Submitted by

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01. Introduction

Rainfall is an important factor affecting agriculture, water resources and the environment. Analysing historical rainfall data helps us understand how rainfall differs across regions, months and years. This project uses Python to analyse historical rainfall records of India covering 1901 to 2015. It includes data cleaning, numerical calculations, statistical analysis, data visualization and web scraping. The results are also presented in an interactive website (dashboard) built from the notebook's logic.

02. Problem Statement and Objectives

The aim of this project is to study rainfall patterns in India and apply core Python libraries in a practical, data-driven way.

- Analyse historical rainfall data of India.
- Clean and process the rainfall data using Pandas.
- Perform numerical calculations using NumPy.
- Compare rainfall across months, years and subdivisions.
- Represent rainfall patterns using charts and graphs.
- Retrieve rainfall-related information using web scraping.

03. Tools and Technologies

Tool / library	Use in the project
Python 3 (Google Colab)	Programming language and notebook environment.
Pandas	Loading the CSV, inspecting, filtering, sorting, grouping and categorising data.
NumPy	Array creation and calculations on monthly rainfall.
SciPy	Pearson correlation between June and July rainfall.
Matplotlib	Bar chart, line chart, histogram and scatter plot.
Seaborn	Boxplot and correlation heatmap.
Requests, BeautifulSoup	Fetching a webpage and extracting its title, headings, links and tables.
HTML, CSS, JavaScript	Interactive project website with the embedded dataset.

04. Dataset Description

Dataset: Rainfall in India (1901–2015), Kaggle, by rajanand

Link: https://www.kaggle.com/datasets/rajanand/rainfall-in-india

The file contains rainfall records (in mm) for each meteorological subdivision of India and each year. Every row holds the monthly rainfall (JAN–DEC), the annual total and four seasonal totals.

Property	Value
Rows (subdivision–year records)	4,116
Columns	19
Unique subdivisions	36
Years covered	1901 – 2015
Columns	SUBDIVISION, YEAR, JAN … DEC, ANNUAL, Jan-Feb, Mar-May, Jun-Sep, Oct-Dec
Average annual rainfall	1,411 mm

Missing values

Most columns contain a few missing values, with ANNUAL having the most. These are skipped during calculations (using `pd.notna`, `dropna`) rather than filled in.

JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC	ANNUAL
4	3	6	4	3	5	7	4	6	7	11	10	26

05. Analysis Performed

The notebook uses the record of Andaman & Nicobar Islands, 1901, to demonstrate basic operations, and then analyses the whole dataset.

5.1 Python basics on a single record

Operation	Result
Rainfall category (Low < 750 mm, Moderate < 1500 mm, High ≥ 1500 mm)	3,373.2 mm → High
Months with at least 100 mm rainfall (loop)	7
Sum of the twelve monthly values	3,373.2 mm
String operations on subdivision name	Length 25; words: ANDAMAN, &, NICOBAR, ISLANDS
List / tuple / set / dictionary	Monthly list; tuple (1901, 3373.2); set of 36 subdivisions; month→rainfall dictionary

5.2 NumPy operations on monthly rainfall

Operation	Result
Array shape / dimensions / dtype	(12,) / 1 / float64

Maximum and minimum monthly rainfall	558.2 mm (Nov) and 2.3 mm (Apr)
Total and average	3,373.2 mm and 281.1 mm
Sorted values	2.3, 29.2, 33.6, 49.2, 87.1, 332.6, 365.1, 388.5, 481.1, 517.5, 528.8, 558.2
Add 10 mm / multiply by 2	Applied element-wise to all 12 months
Pearson correlation, June vs July (all records)	r = 0.7413, p-value ≈ 0

5.3 Pandas analysis on the full dataset

Question	Result
Highest annual rainfall	Arunachal Pradesh, 1948 — 6,331.1 mm
Average annual rainfall	1,411.01 mm

Top five records by annual rainfall

Subdivision	Year	Annual (mm)
Arunachal Pradesh	1948	6,331.1
Arunachal Pradesh	1938	6,129.0
Arunachal Pradesh	1921	5,691.4
Coastal Karnataka	1961	5,553.9
Arunachal Pradesh	1918	5,486.3

A new column, `Rainfall_Category`, was added to every record: Low (< 750 mm), Moderate (750–1499 mm), High (≥ 1500 mm) or Unknown (missing annual value).

06. Data Visualizations

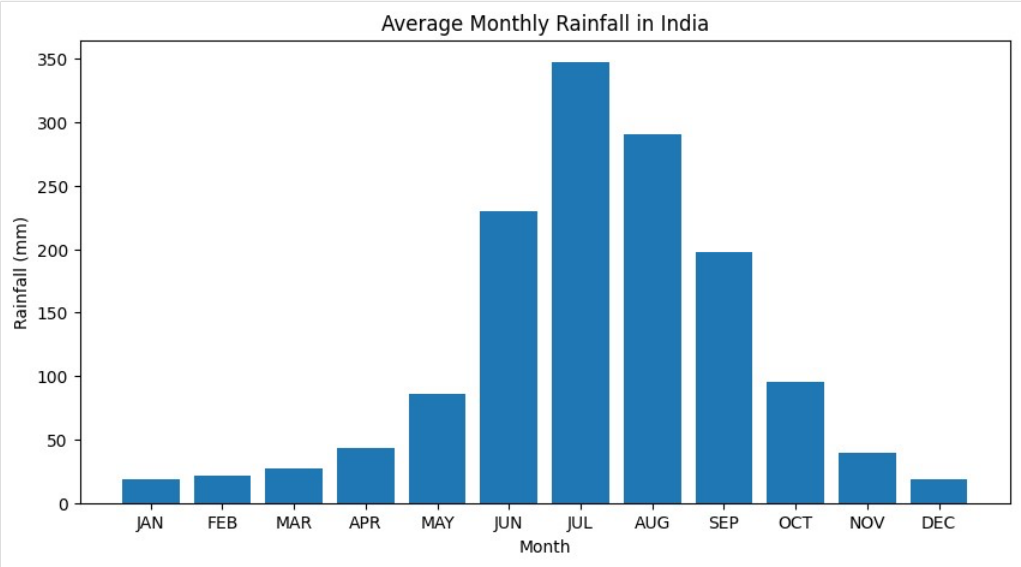


Figure 1. Average monthly rainfall in India. July is the wettest month, followed by August and June.

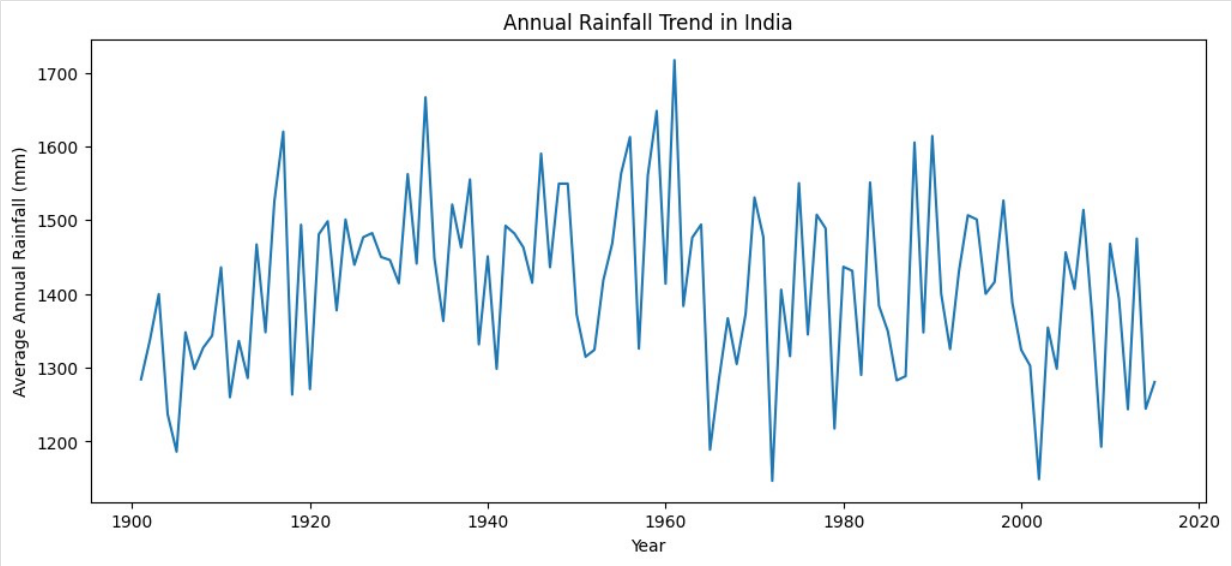


Figure 2. Annual rainfall trend, 1901–2015. Yearly averages move between about 1,150 mm and 1,700 mm with no single steep trend.

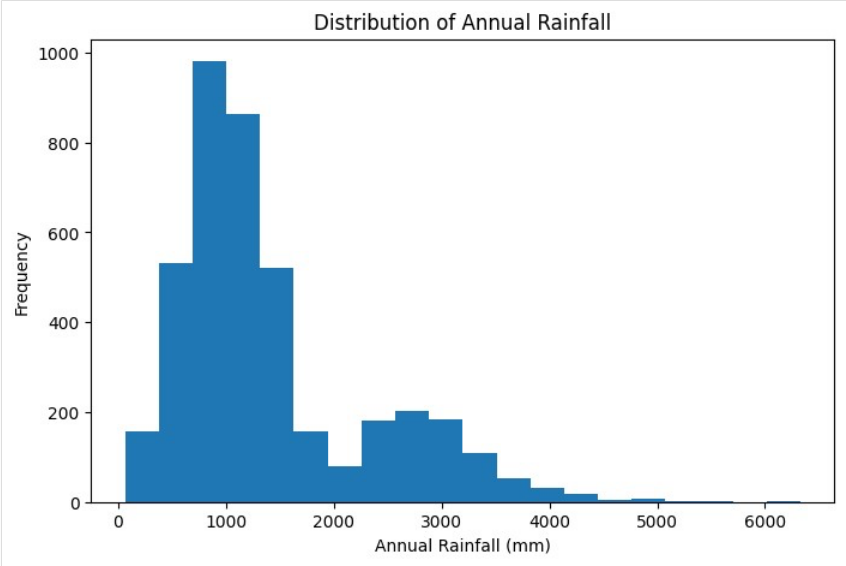


Figure 3. Distribution of annual rainfall across all records.

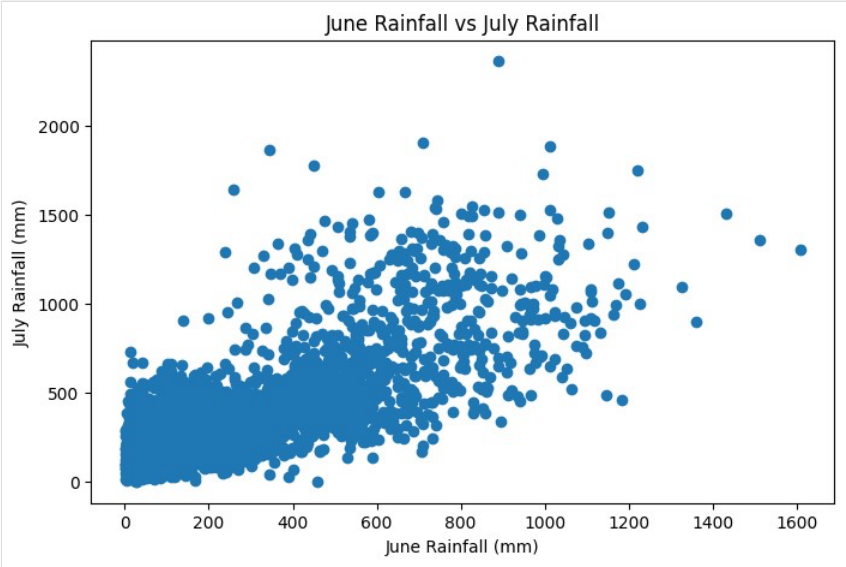


Figure 4. June vs July rainfall (scatter plot); the positive relationship matches r ≈ 0.74.

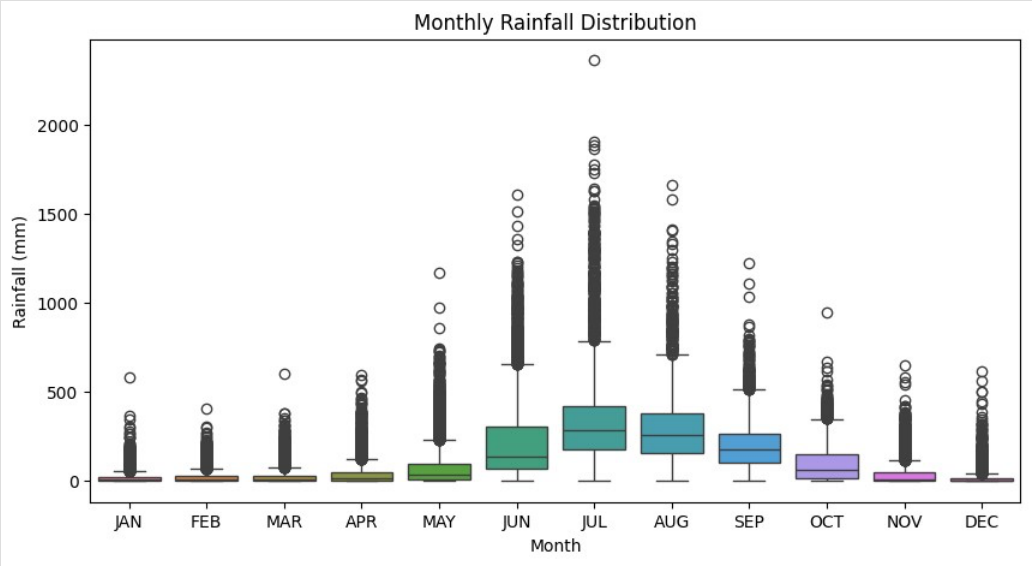


Figure 5. Monthly rainfall distribution (boxplot). Monsoon months have the highest medians and widest spread.

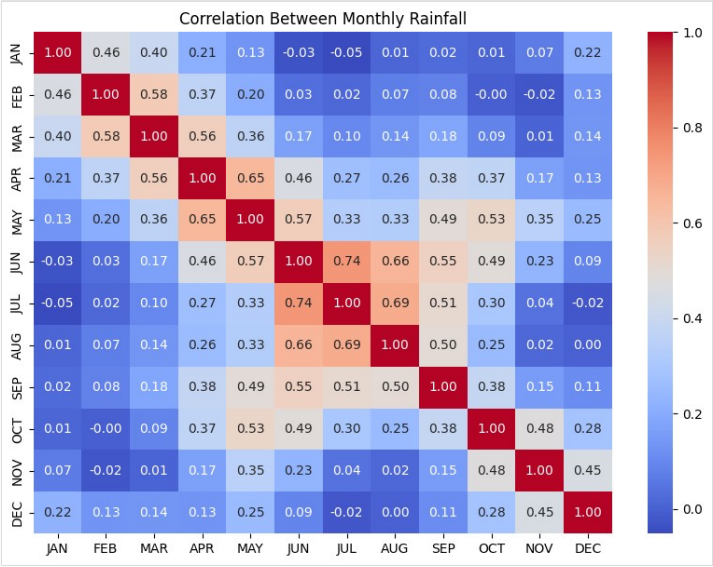


Figure 6. Correlation between monthly rainfall values (heatmap).

07. Web Scraping

Requests and BeautifulSoup were used to open a rainfall-related webpage, read its HTML and extract the title, headings, links and tables.

Step	Result
Page requested	https://iced.niti.gov.in/climate-and-environment/climatevariability/rainfall
HTTP status code	200 (page opened successfully)
Title extracted	Rainfall in India | Kaggle

Headings (h1–h3), links, tables	None found in the downloaded HTML (0 tables)

The downloaded HTML contained no headings, links or tables, which suggests the page builds its content dynamically (for example with JavaScript). Scraping such pages would need a browser-automation tool such as Selenium.

08. Project Website (Dashboard)

An interactive website, *Historical Rainfall Patterns in India (1901–2015)*, was built with HTML, CSS and JavaScript, with the full 4,116-record dataset embedded in the page. It contains:

- **Overview:** key numbers (4,116 rows, 19 columns, 36 subdivisions, 1,411 mm average) and objectives.
- **Quick demo:** live versions of the notebook logic — monthly rainfall bars with an adjustable threshold, rainfall category classifier, and NumPy-style array operations.
- **Live dataset explorer:** choose a subdivision and year range to see average monthly rainfall, annual trend with a 10-year average, category share, comparison of all subdivisions, and wettest and driest years.
- **Analysis walkthrough:** code and results of each practical, in tabs for data exploration, Python structures, NumPy and Pandas.
- **Charts and web scraping:** the six notebook charts (click to enlarge) and the scraping steps.
- Light and dark theme toggle.

09. Conclusion

This project analysed 115 years of rainfall records across 36 subdivisions of India using Python. Pandas was used for cleaning and exploring the data, NumPy for numerical calculations and Matplotlib and Seaborn for visualization. The results show that rainfall is strongly seasonal: June to September contribute most of the annual total, with July the wettest month, and June and July rainfall are strongly correlated ($r = 0.74$). Rainfall also differs greatly between regions; the five highest annual records all come from Arunachal Pradesh and Coastal Karnataka, with a maximum of 6,331.1 mm. The web scraping part showed how a webpage can be fetched and parsed, and its limits for dynamically loaded pages. The interactive website makes the same analysis easy to explore without running any code.